SOFTWARE REQUIREMENTS SPECIFICATION

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1. Introduction

1.1 Project Title

AI Academic Mentor — Personalized Learning & Study Assistant

1.2 Project Overview

AI Academic Mentor is a web-based intelligent academic support platform developed for undergraduate students of Northern University of Business and Technology Khulna (NUBTK). The system leverages Google Gemini AI to provide personalized, on-demand academic assistance, replacing the need for physical tutoring sessions with a scalable, always-available digital alternative.

The platform encompasses six integrated functional modules — AI Chatbot, Quiz Generator, Note Summarizer, Study Planner, Progress Analytics, and Administrative Panel — all delivered through a responsive single-page web application.

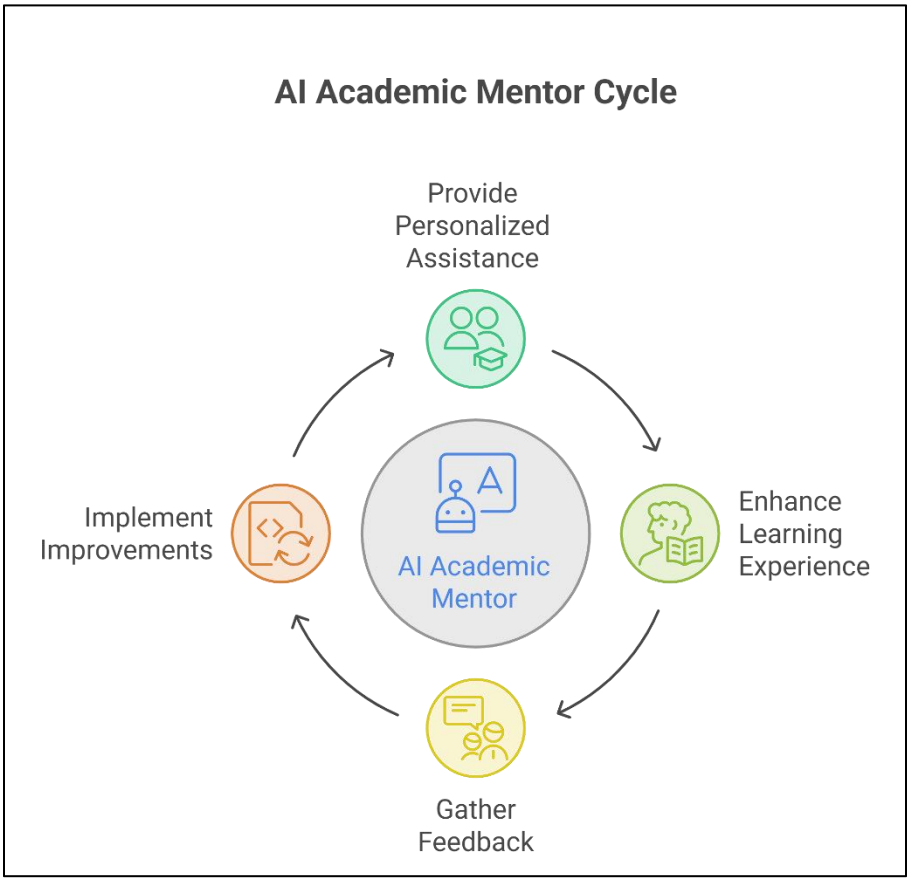


Fig: AI Academic Mentor Cycle

1.3 Problem Statement

Students at NUBTK frequently lack access to academic support outside classroom hours. Physical tutoring is limited in availability, and generic online resources fail to provide personalized guidance aligned with specific course content and individual learning pace. An intelligent, always-available platform serving as a personal academic mentor is needed.

1.4 Objectives

- To provide students with 24/7 AI-powered academic assistance through natural language conversation.
- To automate the generation of subject-specific quizzes and assessment tools.
- To enable intelligent summarization of academic notes and study materials.
- To offer structured, AI-generated study schedules tailored to individual learning goals.
- To track student performance over time and surface actionable insights.
- To provide administrators with oversight tools for managing users and monitoring platform usage.

1.5 Scope of the Project

The system operates entirely within a web browser environment and requires no native application installation. In its current phase, the frontend communicates directly with the Google Gemini API using a user-supplied key, whilst the backend Express.js server provides a production-grade REST API for deployment scenarios requiring centralized data persistence via MongoDB.

The scope of this document is limited to the functional and non-functional requirements of the Version 1.0 release, encompassing all features listed in the project proposal submitted under CSE 4104.

1.6 Overall System Description

Module	Description
AI Chatbot	Multi-turn conversational AI assistant powered by Gemini, with session history persistence.
Quiz Generator	AI-generated subject quizzes (MCQ, True/False, Mixed) with instant scoring and explanations.
Note Summarizer	Paste or type notes for AI-powered summarization with multiple style options.
Study Planner	AI-generated weekly study plans with topic scheduling and PDF export.
Progress Analytics	Dashboard displaying quiz scores, activity heatmaps, and topic-level performance.

1.7 Operating Environment

- Client: Any modern web browser (Chrome 110+, Firefox 110+, Safari 16+, Edge 110+).
- Server: Node.js 18+ with Express.js on any Linux/Windows/macOS host.
- Database: MongoDB 6.0+ via MongoDB Atlas (optional; system degrades gracefully to in-memory mode).
- AI Provider: Google Gemini 2.0 Flash via REST API.
- Network: HTTPS required in production; HTTP acceptable in local development.

1.8 Assumptions and Dependencies

- Users possess a valid Google Gemini API key for AI-powered features.
- The deployment environment has outbound internet access to reach the Gemini API.
- Users operate on devices with a minimum screen width of 320px.
- Passwords are stored with base64 obfuscation in demo mode; bcrypt hashing is enforced in production via the backend.

2. Functional Requirements

Each requirement is assigned a unique identifier (FR-XXX), a descriptive name, a rationale, and a priority classification (High / Medium / Low).

ID	Requirement	Description	Priority
Authentication & User Management			
FR-01	User Registration	A new user shall be able to create an account by providing name, email, student ID, department, and a password of at least six characters.	**High**
FR-02	User Login	Registered users shall authenticate using email and password, receiving a persistent session token upon success.	**High**
FR-03	Demo Accounts	The system shall pre-seed two demo accounts (student and admin) to enable immediate evaluation without registration.	**Medium**
FR-04	Session Persistence	Authenticated sessions shall persist across page reloads via MongoDB Atlas -based token management.	**High**
FR-05	Logout	Users shall be able to terminate their session from any page via the sidebar user profile widget.	**High**
FR-06	Update Profile	Users can update their profiles including name, student ID, and department via the Settings page.	**Medium**
AI Chatbot			
FR-07	Multi-Turn Conversation	Users shall engage in multi-turn academic conversations with the AI, with the full message history passed on each request for contextual coherence.	**High**
FR-08	Session Management	Users shall be able to create, switch between, and delete named conversation sessions.	**Medium**
FR-09	Conversation Persistence	All conversation sessions shall be saved to MongoDB Atlas and restored on subsequent visits.	**Medium**

ID	Requirement	Description	Priority
Quiz Generator			
FR-10	Topic-Based Quiz Generation	Users shall input a topic, question count (1–20), difficulty (Easy/Medium/Hard), and question type (MCQ/True-False/Mixed) to generate a quiz.	**High**
FR-11	Interactive Quiz Taking	Generated quizzes shall be presented one question at a time with immediate answer validation and explanations.	**High**
FR-12	Quiz Result Storage	Completed quiz results (topic, score, question count, date) shall be stored and reflected in Progress Analytics.	**High**
Note Summarizer			
FR-13	Note Summarization	Users shall paste or type academic notes for AI-powered summarization using their preferred style (Concise/Detailed/Bullet Points/Key Concepts).	**High**
FR-14	Summarization History	The system shall retain a history of all summaries within a user session.	**Medium**
Study Planner			
FR-15	AI Study Plan Generation	Users shall input a topic, available days per week, daily hours, and target date to receive an AI-generated structured weekly study plan.	**High**
FR-16	Plan Persistence	Generated study plans shall be saved to MongoDB Atlas for future reference.	**Medium**
Progress Analytics			
FR-17	Performance Dashboard	The system shall display aggregate statistics including total quizzes taken, average score, best score, and activity streak.	**High**
FR-18	Score Trend Chart	The system shall render a bar chart of the last ten quiz scores with color-coded performance bands.	**Medium**
FR-19	Topic Breakdown	The system shall aggregate quiz results by topic and display average performance per topic.	**Medium**
FR-20	Activity Heatmap	The system shall display weekly activity distribution across all platform features.	**Low**

ID	Requirement	Description	Priority
Administration			
FR-21	User Listing	Administrators shall view a list of all registered users with their name, email, student ID, department, and join date.	**High**
FR-22	Admin Access Control	The admin panel shall be accessible only to users with the role attribute set to 'admin'.	**High**
FR-23	Data Reset	Users shall be able to clear all personal data from the platform via the Settings page.	**Medium**

3. Non-Functional Requirements

ID	Requirement Description	Priority	Rationale
Performance			
NFR-01	AI API responses shall be initiated within 300 ms of user submission, with full streaming completion within 3–10 seconds.	**High**	Ensures responsive AI interaction experience.
NFR-02	Page transitions and UI interactions shall complete within 100 ms on modern hardware.	**High**	Maintains fluid, lag-free navigation.
NFR-03	The backend API shall support a minimum of 100 concurrent users under standard load via express-rate-limit middleware.	**High**	Prevents server overload under real usage.
Usability			
NFR-04	The application shall be fully responsive across screen widths from 320px (mobile) to 2560px (widescreen).	**High**	Supports diverse student device types.
NFR-05	Navigation shall be achievable in three clicks or fewer from any page to any core feature.	**Medium**	Reduces cognitive load and improves UX.
NFR-06	All form validation errors shall be displayed in line with explanatory messages.	**Medium**	Guides users on input correction immediately.

ID	Requirement Description	Priority	Rationale
NFR-07	The interface shall conform to WCAG 2.1 Level AA contrast ratios for accessibility.	Medium	Ensures inclusivity for visually impaired users.
Security			
NFR-08	All JWT tokens shall be signed with a minimum 64-character random secret in production.	High	Prevents token forgery and session hijacking.
NFR-09	Passwords shall be hashed using bcrypt with a cost factor of 12 before database storage.	High	Protects credentials against brute-force attacks.
NFR-10	The backend shall apply HTTP security headers via the helmet middleware.	High	Mitigates common web attack vectors (XSS, CSRF).
NFR-11	The Gemini API key shall never be exposed in server-rendered HTML or version control.	Critical	Prevents API key theft and unauthorized usage.
NFR-12	CORS shall be restricted to the configured CLIENT_URL origin in production environments.	High	Blocks unauthorized cross-origin API access.
NFR-13	AI request rate limiting shall be enforced at both the backend (20 req/min) and frontend levels.	High	Prevents API abuse and cost overruns.
Reliability			
NFR-14	The system shall degrade gracefully to a fully functional demo mode when no MongoDB connection is available.	Medium	Ensures continuity of service during DB outages.
NFR-15	All MongoDB Atlas operations shall be wrapped in try-catch blocks to prevent data corruption from JSON parse errors.	High	Maintains data integrity at all times.
NFR-16	The backend shall return structured JSON error responses for all 4xx and 5xx scenarios.	Medium	Enables consistent client-side error handling.
Maintainability			
NFR-17	Frontend code shall be modularised into reusable components (auth.js, sidebar.js) shared across all pages.	Medium	Reduces duplication and eases future updates.
NFR-18	Backend routes shall be separated by functional domain (auth, ai, quiz, plan, summary, progress, admin).	Medium	Improves codebase readability and modularity.

ID	Requirement Description	Priority	Rationale
NFR-19	Environment-specific configuration shall be managed exclusively through .env files, never hardcoded.	**High**	Simplifies deployment across environments.
Scalability			
NFR-20	The stateless REST API design permits horizontal scaling behind a load balancer without session affinity.	**Medium**	Supports growth in concurrent user base.
NFR-21	Database queries shall use indexed fields (email, _id) to maintain sub-10ms response times at scale.	**High**	Prevents DB bottlenecks as data volume grows.

4. User Roles and Permissions

The system identifies three user classes with distinct permissions and responsibilities.

4.1 Student / User

- Register an account and log in securely.
- Interact with all AI-powered modules (Chatbot, Quiz Generator, Note Summarizer, Study Planner).
- View personal progress analytics and performance metrics.
- Manage personal data, settings, and session history.

4.2 Administrator

- Access the Admin Panel to view all registered users.
- Monitor platform statistics and activity overview.
- Manage user accounts and oversee platform health.
- Access all features available to students.

4.3 Guest (Unauthenticated Visitor)

- View the public landing page and project information.
- Access the login and registration pages.
- Cannot access any authenticated feature without logging in.

Feature	Guest	Student	Administrator
Landing Page / Login	**Yes**	**Yes**	**Yes**
Registration	**Yes**	—	—
AI Chatbot	**No**	**Yes**	**Yes**
Quiz Generator	**No**	**Yes**	**Yes**
Note Summarizer	**No**	**Yes**	**Yes**
Study Planner	**No**	**Yes**	**Yes**
Progress Analytics	**No**	**Personal only**	**All users**
Settings / Data Reset	**No**	**Yes**	**Yes**
Admin Panel	**No**	**No**	**Yes**
User Management	**No**	**No**	**Yes**

5. Use Cases

UC-01: User Registration

Use Case ID	UC-01
Name	User Registration
Actor	Guest (unauthenticated visitor)
Precondition	The user has navigated to the registration page and has not previously registered.
Main Flow	1. User enters name, email, student ID, department, and password. 2. System validates all fields and checks email uniqueness. 3. System creates the account, saves to MongoDB Atlas, and redirects to Dashboard.
Alternate Flow	If the email already exists, the system displays 'An account with this email already exists' and halts.
Postcondition	A new user account is created; the user is authenticated and redirected to Dashboard.

UC-02: User Login

Use Case ID	UC-02
Name	User Login
Actor	Student / Administrator
Precondition	User has a registered account.
Main Flow	1. User enters email and password on the login page. 2. System validates credentials against the stored bcrypt hash. 3. System issues a signed JWT session token and redirects to Dashboard.
Alternate Flow	If credentials are invalid, the system displays an authentication error message and the form remains active.
Postcondition	User is authenticated and granted access to all features permitted by their role.

UC-03: AI Chatbot Interaction

Use Case ID	UC-03
Name	AI Chatbot Interaction
Actor	Student
Precondition	User is authenticated and has configured a Gemini API key in Settings.
Main Flow	1. User types an academic question and submits. 2. System appends the message to the conversation history. 3. System sends the full history to Gemini API and awaits response. 4. System displays the AI reply in the chat interface. 5. Conversation is saved to the current session.
Alternate Flow	If no API key is configured, the system prompts the user to enter one before proceeding.
Postcondition	The conversation is persisted to MongoDB Atlas and visible in the session list.

UC-04: Quiz Generation and Assessment

Use Case ID	UC-04
Name	Quiz Generation and Assessment
Actor	Student
Precondition	User is authenticated and has a Gemini API key configured.
Main Flow	1. User selects topic, question count, difficulty, and type. 2. System sends a structured prompt to Gemini and parses the JSON response. 3. System renders the quiz one question at a time. 4. On completion, system calculates and displays the score. 5. Results are saved to Progress Analytics.
Alternate Flow	If the AI response cannot be parsed as valid JSON, the system displays an error and offers a retry.
Postcondition	Quiz result (topic, score, count, date) is saved and reflected in the Progress Analytics dashboard.

UC-05: Note Summarization

Use Case ID	UC-05
Name	Note Summarization
Actor	Student
Precondition	User is authenticated and has a Gemini API key configured.
Main Flow	1. User pastes or types academic notes into the Note Summarizer input area. 2. User selects a preferred summarization style (Concise / Detailed / Bullet Points / Key Concepts). 3. System sends the notes and style preference to Gemini API. 4. System displays the AI-generated summary. 5. Summary is retained in session history.
Alternate Flow	If the input notes are empty or below the minimum length, the system displays a validation error and prevents submission.
Postcondition	The generated summary is displayed and stored in the session's summarization history for future reference.

UC-06: Study Plan Generation

Use Case ID	UC-06
Name	Study Plan Generation
Actor	Student
Precondition	User is authenticated and has a Gemini API key configured.
Main Flow	1. User inputs topic, days/week, hours/day, and target date. 2. System generates a structured weekly plan via Gemini. 3. System displays the plan and saves it to MongoDB Atlas.
Alternate Flow	If the Gemini API returns an error, the system shows the error message and allows the user to retry.
Postcondition	A named study plan is saved and accessible from the Planner page history.

UC-07: Admin User Management

Use Case ID	UC-07
Name	Admin User Management
Actor	Administrator
Precondition	User is authenticated with the 'admin' role.
Main Flow	1. Admin navigates to the Admin Panel. 2. System loads the registered user list from MongoDB Atlas. 3. Admin views names, emails, student IDs, departments, and roles.
Alternate Flow	If no users have registered beyond the demo accounts, the system displays only demo account records.
Postcondition	Administrator has reviewed the user list; no data is modified.